Highlights

Collusion-resilient, tamper-evident range analytics over homomorphically encrypted Internet of Things data in multi-cloud storage

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- Range aggregates over homomorphically encrypted IoT data held by many clouds
- Query, matched records and result stay hidden while one cloud is honest
- Encrypted summed-area tables turn any d-dimensional box into 2^d point keys
- A fresh secret scalar per query makes any tampered answer evident
- Real-data range queries take under 1.2 s per cloud with one 132 kB reply

Collusion-resilient, tamper-evident range analytics over homomorphically encrypted Internet of Things data in multi-cloud storage

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ABSTRACT

Internet of Things (IoT) data are increasingly stored encrypted and replicated across several clouds, where analysts query aggregates over time, place and sensor-value ranges. Existing designs fall short: hiding which records a query selects is undone once the clouds aggregate over them, and prior multi-server systems compute over plaintext, need an honest majority with server interaction, or leak access patterns, and cannot cheaply detect tampering by colluding clouds. We propose CR-SHE (Collusion-Resilient Searchable Homomorphic Encryption), which combines distributed point functions with lattice-based homomorphic encryption: each cloud contracts its share of a hidden selection against owner-encrypted columns and returns one ciphertext. Our contributions are one-round multi-dimensional range aggregation over encrypted summed-area tables, where any box becomes 2^d padded point keys; tamper evidence from a secret scalar drawn per query inside the keys, accepting a wrong answer with probability about 2^{-36} at no extra homomorphic cost; role-scoped access without interaction; and security proofs against $n-1$ colluding clouds. On two real IoT datasets (825,554 records), our implementation answers a verified range query in 0.68–1.19 s per cloud with one 132 kB ciphertext, flat from one minute to the whole timeline, uploading 5.5 kB against 44 MB for a single-server encrypted query; no tampered answer was accepted in 4,500 trials, and a frequency attack fell to random guessing. CR-SHE lets multi-cloud IoT deployments outsource range analytics without trusting any single cloud with the query, the data or the result.

1. Introduction

Connected sensors, wearables and cyber-physical systems make the Internet of Things (IoT) one of the largest sources of continuously generated data [1]. The people who query these data, among them clinicians, grid operators, city agencies and maintenance contractors, are rarely the ones who own the devices, so the data are served from the cloud, and deployments increasingly replicate them across several providers for capacity, availability and freedom from vendor lock-in [2, 3]. Much of it is sensitive (physiological telemetry, mobility traces, industrial diagnostics), so it must be encrypted before it leaves the IoT trust domain and stay encrypted in the cloud while many stakeholders analyse it.

Encryption at rest takes away what the cloud was for. Searchable encryption lets a provider find records matching an encrypted keyword [4–6], and leveled homomorphic encryption lets it evaluate bounded-depth arithmetic on ciphertexts [7, 8], but in their single-server form both leak the access pattern, which leakage-abuse attacks turn into plaintext recovery [9–11]. Replication offers an escape: if the client splits its query so that no provider sees it, the access pattern stays hidden as long as one provider abstains from collusion. Splinter [12] established this model with function secret sharing, DORY [13] brought it to encrypted documents, and Waldo [14] to predicates over time series.

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Hiding the selection is worthless if the computation gives it back. A scheme can privately determine which records match a query and then, to aggregate over them, hand the matched set to the providers, at which point the coalition learns the access pattern exactly. We call this the *selection-to-computation composition leak*. Existing distributed-trust systems avoid it by not computing at all (DORY), by computing over data the servers can read (Splinter), by an honest-majority setting with server interaction (Waldo), or by Shamir-shared data whose share multiplications need $2l + 1$ servers for a one-round predicate of length l and an honest majority (Obscure [15]). Multi-cloud IoT deployments add three constraints: (i) privacy must hold while any one provider is honest, $t = n - 1$; (ii) providers never talk to one another; and (iii) the data are encrypted once under the owner's key, so a replica can be added, lost or revoked by copying ciphertexts. And IoT analytics are *range* queries: a clinician asks for one patient's mean heart rate across a night, a city agency for the hours carbon monoxide sat in its top bands at one site. Each is an aggregate over a box in a few ordered dimensions, which single-keyword aggregates answer only with one query per tag in the box, revealing its size.

We propose CR-SHE (Collusion-Resilient Searchable Homomorphic Encryption). The data owner replicates a one-time-padded index and homomorphically encrypted columns to n clouds. A query is split into distributed point function (DPF) keys, one per cloud for a keyword and 2^d for a d-dimensional range; each cloud contracts its *own share* of the hidden selection against the encrypted columns and returns one ciphertext, and the client adds the n ciphertexts and decrypts once. The matched set is never assembled at any cloud, and its size never leaves the client. The main contributions of this paper are as follows:

- **One-round oblivious range aggregation.** The owner stores encrypted summed-area tables of registered data cubes, and any d-dimensional box becomes 2^d DPF keys with signed outputs at its corners, padded with dummy keys so that boxes at the cube's edge look like interior ones. Provider work and the response depend neither on the range nor on the record count.
- **Tamper evidence from a per-query secret.** A second key set carries a scalar δ_s drawn afresh for each query, so the client accepts a wrong answer with probability at most $1/(p - 1)$ plus the DPF advantage, whatever up to $n - 1$ deviating clouds compute, with no stored key and no extra homomorphic product.
- **Role-scoped access without interaction.** Role-filtered columns and role-specific mask keys confine each user to its own role's records, without any exchange between providers.
- **Security analysis and implementation on real IoT data.** We prove leakage security against $n - 1$ colluding semi-honest clouds and show why a homomorphic layer is needed at $t = n - 1$ without interaction. A C++ prototype and an OpenFHE-based reference implementation are evaluated on two real IoT datasets with 825,554 records, covering range queries, injected tampering, role confinement and a frequency attack against padding baselines.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the system model, the threat model and the cryptographic preliminaries. Section 4 describes the CR-SHE construction, and Section 5 analyses its security. Section 6 describes the implementation and the experimental setup, and Section 7 presents and discusses the results. Finally, Section 8 concludes the paper. Full proofs and additional experiments are provided in the Supplementary Material.

2. Related work

Distributed-trust query systems. Splinter [12] evaluates function-secret-sharing queries against replicas of *public* cleartext data, with sums, counts and top-k under equality and one-interval conditions, honest-but-curious providers and no protection against incorrect results. DORY [13] searches encrypted documents: replicas hold a Bloom-filter index one-time-padded under a per-folder key, hiding access patterns and result counts from $\ell - 1$ of ℓ malicious trust domains, with security with abort through aggregate MACs; it stops at retrieval. Waldo [14] filters secret-shared time series with malicious security in a three-party honest-majority setting with server interaction. Obscure [15, 16] answers count, sum, average, maximum and top-k queries over Shamir-shared unary-encoded data at non-communicating servers with information-theoretic security

Table 1
Comparison with related systems (✓ supported, ○ partial, × not supported).

System	Data at servers	Oblivious analytics	Range aggregates	Collusion bound	Server channel	Tamper evidence
Splinter [12]	plaintext replica	○ (plaintext)	○ (1-D)	$n-1$	no	×
DORY [13]	encrypted replica	×	×	$n-1$	no	abort
Waldo [14]	secret-shared	✓	✓	1 (honest maj.)	yes	✓
Obscure [15]	Shamir-shared	✓	○ (1-D)	minority	no	optional
Hafiz et al. [17]	plaintext replica	○ (plaintext)	×	$\ell-1$	no	○ (robust)
HADES [22]	plaintext, 1 server	○ (plaintext)	✓	n/a	n/a	×
ARQ [21]	encrypted, 1 server	×	✓ (leaky)	n/a	n/a	×
Timecrypt, Zeph [23, 24]	encrypted, 1 server	✓	time, revealed	n/a	n/a	×
Multi-cloud SSE [38]	encrypted	×	×	×	no	×
CR-SHE (this work)	encrypted replica	✓	✓, 1 round	$n-1$	no	✓ (abort)

and optional verification; its adversary may not control a majority of the servers, a one-round predicate of length l needs $2l+1$ servers, and ranges become point queries or nodes of a precomputed tree, measured in one dimension. Hafiz et al. [17] answer aggregate queries in one round with information-theoretic PIR over plaintext database copies, private against $\ell-1$ of ℓ servers, without range predicates. CR-SHE differs from all of them in combining $t=n-1$ with $n=2$ sufficing, owner-encrypted replicas, one-round d-dimensional box aggregates and tamper evidence.

Private range aggregation. Prefix sums answer a range sum from a constant number of lookups, as in OLAP data cubes [18] and summed-area tables [19]. Boyle, Gilboa and Ishai give function secret sharing for d-dimensional intervals through the same inclusion–exclusion over 2^d anchored intervals [20, Cor. 3.4]; we apply that decomposition on the data side, which hides record coordinates that interval keys would need in the clear. Over encrypted data at one server, ARQ [21] builds range aggregates from structured encryption with query-dependent leakage, and HADES [22] evaluates encrypted range predicates against a single provider's *plaintext* database in one round, without result verification; it reports that one-round homomorphic processing of encrypted data takes more than fifteen hours for a million records. Timecrypt [23] and Zeph [24] aggregate encrypted time series but reveal the queried interval.

Verification and access control. Verifiable DPFs [25] and access control for FSS [26, 27] guard against a malicious *client*. Against malicious servers, verifiable homomorphic secret sharing [28] checks evaluation, and PVODTE [29] has the client send a one-time MAC key under homomorphic secret sharing to two servers, secure against one malicious server. Our check uses the same kind of information-theoretic MAC, y alongside $\delta_s y$ as in SPDZ [30], but carries δ_s inside the DPF outputs, needing no public-key operation, and covers $n-1$ colluding providers.

PIR and encrypted search. DPF-based selection is multi-server PIR [31]; its linear server work is the price of PIR without preprocessing [32], which preprocessing PIR avoids with client hints that updates invalidate [33]. Single-server PIR from lattices [34, 35] removes the non-collusion assumption at a large uplink cost, which we measure as a baseline. Volume-hiding structured encryption [36, 37] pads posting lists, costly on heavy-tailed IoT data; multi-cloud searchable encryption [38, 39] and multi-cloud homomorphic computation [40] do not hide the access pattern from colluding clouds. Table 1 summarises the comparison.

3. System model, threat model and preliminaries

Entities. The **data owner** (DO), an IoT-domain authority such as a hospital or city data office, holds the tag dictionary D, the mask key K' and the homomorphic secret key sk. Producers and edge relays forward records and hold no secrets. The n **cloud providers** $\mathrm{CP}_1,\ldots,\mathrm{CP}_n$ each hold the public key pk and a replica of the masked index and of the encrypted columns and tables, and never decrypt. Authorised **data users** (DU) hold D, their role's mask key and delegated decryption, issue queries and combine responses. In Fig. 1, the owner replicates a masked index and encrypted columns to n clouds (1); a query is split into DPF keys, one set per cloud (2); each cloud aggregates obliviously over its own share (3) and returns one ciphertext carrying the value and a check value (4); and the user combines, verifies and decrypts (5). No cloud sees the matched set or its size. For example, a city's monitoring stations feed its data office, which replicates the encrypted store to three clouds; a health agency asks for the hours carbon monoxide spent

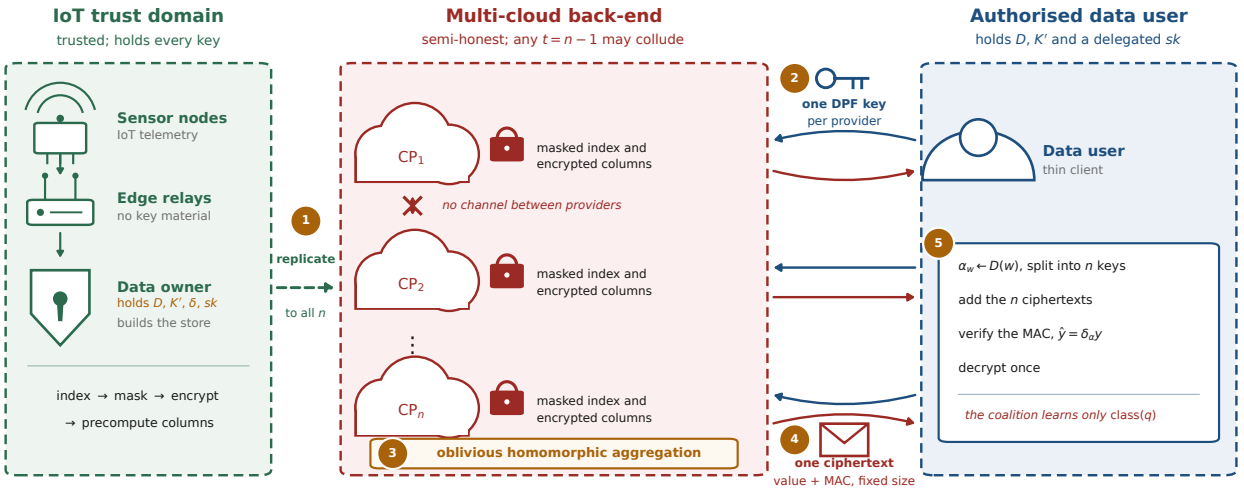


Figure 1: CR-SHE system architecture. Drawn with an AI-assisted script.

in its top bands at one site last month, and no cloud learns the site, the month, the bands, the number of matching hours or the answer.

Threat model. The adversary is probabilistic polynomial-time, static and semi-honest: it corrupts a coalition C of at most $t = n - 1$ providers, which follow the protocol and pool everything they see, over adaptively chosen queries. The goal is that this view be simulatable from data-independent leakage that reveals neither the keyword or range, the access pattern, the result-set size nor the result. The DO and authorised DUs are honest, channels are authenticated and confidential, and side channels beyond message sizes are out of scope. Beyond this goal we address two active threats: *deviating providers*, which may return any response at all (Section 4.6), and *users exceeding their authority*, who may send malformed keys or arbitrary share vectors (Section 4.7). Collusion between a DU and a provider is excluded, since the DU already holds decryption capability.

Primitives. A pseudorandom function (PRF) F has distinguishing advantage $\mathrm{Adv}^{\mathrm{prf}}$ against a random function [41]. An n-party DPF [20, 42, 43] for point functions $f_{\alpha,\beta} : [N] \to \mathbb{Z}_p$ is a pair $(\mathsf{Gen}, \mathsf{Eval})$ with correctness $\sum_j \mathsf{Eval}(j, \kappa_j, x) = f_{\alpha,\beta}(x)$ and $(n-1)$-privacy: the keys of any $n - 1$ parties are simulatable without (α, β), up to $\mathrm{Adv}^{\mathrm{dpf}}$. The two-party construction has keys of $O(\lambda \log N)$ bits and full-domain evaluation in $O(N)$ PRG calls; for $n > 2$ we use the additive sharing of a unit vector in which $n - 1$ parties hold seeds and one an explicit vector. Leveled BFV [7, 8] over plaintext ring $\mathbb{Z}_p$ is IND-CPA with advantage $\mathrm{Adv}^{\mathrm{cpa}}$; we write $\boxplus$ for ciphertext addition and $\boxdot$ for plaintext-ciphertext multiplication, and use ℓ SIMD slots arranged in two rows of $\ell/2$ [44]. Table 2 lists the notation.

4. The CR-SHE construction

4.1. Overview, and why a homomorphic layer

The DO builds a membership index $I \in \{0,1\}^{N \times N_d}$ in which keyword w occupies row $\alpha_w = D(w)$ of a secret, append-only dictionary, stores it masked (Section 4.3), and encrypts aggregate columns, cube tables and the record column under pk (Algorithm 1). Rather than recovering the selection and asking providers to aggregate over the named records, which hands them S, each provider contracts its own share of the selection against encrypted data (Algorithm 2).

The obvious alternative, dealing the data as an (n, n) additive sharing and answering DPF queries over the shares, fails under our constraints.

Table 2
Notation.

Symbol	Meaning
$n,\ t$	number of providers; collusion bound $t = n - 1$
$N,\ N_d$	tag domain size; number of records
$p,\ \ell$	plaintext modulus and DPF output group; SIMD slots
$I,\ I',\ P$	membership index; masked index; one-time pad
$\alpha_w,\ \kappa_j,\ e_j$	row of keyword w; key of CP_j; its full-domain evaluation
$m_i,\ w_f$	value of record i; weights of template f
A_f	aggregate column of template f over the tag domain
$d,\ M$	cube dimension; number of cells $\prod_k s_k$
$C_f,\ T_f$	cell aggregates; their summed-area table
$\kappa_\sigma(B),\ \beta_\sigma$	corner $\sigma \in \{0,1\}^d$ of box B; its sign or 0
δ_s	per-query secret of the tamper check, uniform on $\mathbb{Z}_p^*$
$\rho,\ K'_\rho$	role; its mask key
$S,\ \lvert S\rvert$	matched record set; its size

Proposition 1 (No local answers over shares). *Let a column $A \in \mathbb{Z}_p^N$ be dealt as $A = \sum_{k\in[n]} S_k$ with $S_1, \ldots, S_{n-1}$ uniform, and let e_j be the evaluation vector of the DPF key κ_j for $f_{\alpha,1}$. If each provider returns $r_j = \langle e_j, S_j\rangle$, then $\sum_j r_j - A[\alpha] = -\sum_{j\neq k}\langle e_j, S_k\rangle$, and for $n = 2$ this error is uniform on $\mathbb{Z}_p$ except with negligible probability.*

Proof. $A[\alpha] = \sum_{j,k}\langle e_j, S_k\rangle$ and the answers supply only the diagonal. For $n = 2$, substituting $S_2 = A - S_1$ gives the error $-\langle e_1, A\rangle + \langle e_1 - e_2, S_1\rangle$, uniform since S_1 is uniform and independent of the keys whenever $e_1 \neq e_2$; $e_1 = e_2$ would let a provider read α from its own key, so it has negligible probability by DPF privacy. □

The cross terms are not an artefact of linear answers. In any non-interactive scheme where a provider holds an additive share, queries are formed without reference to the data, and the scheme is correct on every share tuple, a client holding a database X at a single server can set that server's share to X, play the other providers itself and subtract their contributions; this yields single-server PIR, which implies oblivious transfer [45] and is instantiated by homomorphic encryption [46]. Within that class, sharing the data moves the public-key primitive into a per-share PIR rather than removing it; a lower threshold ($t < n/2$, as in Obscure), interaction or preprocessing escapes the argument (Supplementary S1). Replicas under one owner key also survive the loss of $n - 1$ providers, keep answering over any surviving pair, and let the owner add or revoke a provider by copying ciphertexts.

4.2. Oblivious homomorphic aggregation

Suppose the providers hold additive shares $v_1, \ldots, v_n \in \mathbb{Z}_p^M$ of a vector v and replicas of an encrypted column $\mathrm{C} = (\mathsf{Enc}_{pk}(c_1), \ldots, \mathsf{Enc}_{pk}(c_M))$. Provider j returns

$$\mathrm{ct}_j = \boxplus_{i\in[M]} v_j[i] \boxdot \mathrm{C}_i, \tag{1}$$

and $\boxplus_j \mathrm{ct}_j$ decrypts to $\langle v, c\rangle$. Nothing that leaves a provider depends on v beyond its own share, and the response is one ciphertext of fixed size whatever the support of v.

Lemma 1 (Algebraic compatibility). *If the DPF output group and the plaintext ring are the same $\mathbb{Z}_p$ and $\sum_i |v[i]\, c_i| < p$, then the contraction (1) summed over the providers decrypts to $\sum_i v[i]\, c_i$ over the integers.*

Proof. $\boxdot$ and $\boxplus$ are homomorphisms over the same ring, so summation commutes with decryption; the magnitude bound prevents wraparound. Decryption correctness is a noise-budget question: a plaintext-ciphertext product by a uniform share in $\mathbb{Z}_p$ scales the noise by up to $p \approx 2^{36}$ without consuming a level, which the parameters must absorb; we check the worst case empirically (Section 7.1). □

Instantiating the DPF over $\mathbb{Z}_{2^k}$ and the homomorphic scheme over an unrelated modulus, the natural default when the two libraries are chosen independently, silently breaks correctness through wraparound.

4.3. The masked, packed index

Stored in the clear, the index would hand every provider each posting-list length, which the attack of Section 7.5 turns into tag identities. As DORY does for its Bloom-filter bits [13], the DO stores it one-time-padded, here with field-element masks that arithmetic DPF shares can strip, and packed $\pi = 32$ membership bits per word:

$$I'[x][c] = \sum_{b<\pi} I[x][\pi c + b]\, 2^b + P[x][c] \bmod p, \tag{2}$$

where $P[x][c]$ is drawn from $F'_{K'}(x, c)$ by rejection sampling: a 37-bit candidate is kept only if it is below p, so each mask word is exactly uniform. A packed word is below $2^{32} < p$, so the pad hides it perfectly, and the index shrinks 32-fold; the four spare bits of each word catch a uniformly corrupted word except with probability 1/16.

Theorem 1 (Index hiding). *For any two indexes I_0, I_1, no* PPT *distinguisher tells* $\mathrm{pack}(I_0) + P$ *from* $\mathrm{pack}(I_1) + P$ *with advantage greater than* $\mathrm{Adv}^{\mathrm{prf}}_{F'}$*; the stored index reveals no posting-list length and no membership relation.*

Proof. With F' replaced by a random function, rejection sampling makes every mask word independent and exactly uniform on $\mathbb{Z}_p$, so $\mathrm{pack}(I_b) + P$ is a one-time pad for either b. □

Reducing 64-bit words modulo p instead biases each mask word by $r(p - r)/(p\, 2^{64}) \approx 8.27 \times 10^{-10}$, $r = 2^{64} \bmod p$; over the 146,625,960 words of the packed telemetry index the union bound is about 0.12, so rejection sampling is what makes Theorem 1 hold as stated.

4.4. Single-keyword aggregates and the packed response

For an aggregate over one keyword's posting list under a template registered at setup, the DO precomputes $A_f[x] = \sum_i I[x][i]\, w_f[i]\, m_i$ for every tag x and encrypts it as packed ciphertexts. A query is one DPF key per provider for $f_{\alpha_w,1}$, and (1) with $v_j = e_j$ over A_f costs $O(N)$ PRG calls and $O(N/\ell)$ plaintext-ciphertext products, independent of N_d. The DO places A_f in slot row 0 and its check column in row 1, chunked $\ell/2$ tags per ciphertext; the provider multiplies each chunk by its evaluation vector, adds the chunks and rotates and sums within the rows, so slot 0 carries the value share and slot $\ell/2$ the check share in one ciphertext. With k templates each row is cut into k segments, so one ciphertext answers, for instance, a count and a sum (a mean) or a count, sum and sum of squares (a variance). Modulus switching to one RNS tower halves the response. Conjunctions and weights chosen at query time use a two-round flexible path (Section 4.7).

4.5. Oblivious range aggregation over data cubes

Data cubes. The DO registers a cube of d ordered tag families of sizes $s_1, \ldots, s_d$, for example device, one-minute window and the decile band of one sensor, such that each record i falls in exactly one cell $c(i) \in \prod_k [s_k]$. For each template f it computes the cell aggregates $C_f[c] = \sum_{i:\, c(i)=c} w_f[i]\, m_i$ and the summed-area table $T_f[c] = \sum_{c' \le c} C_f[c']$, with $\le$ coordinate-wise, flattened into a column of $M = \prod_k s_k$ entries and encrypted like A_f.

Queries. For a box $B = \prod_k [lo_k, hi_k]$ and $\sigma \in \{0,1\}^d$, let corner $\kappa_\sigma(B)$ take $lo_k - 1$ on the axes where $\sigma_k = 1$ and hi_k elsewhere. By inclusion–exclusion,

$$\sum_{c \in B} C_f[c] = \sum_{\sigma \in \{0,1\}^d} (-1)^{|\sigma|}\, T_f[\kappa_\sigma(B)], \tag{3}$$

where a corner with a coordinate -1 lies outside the cube and contributes 0. The DU sends, for every σ, one DPF key per provider for the point $\kappa_\sigma(B)$ with output $\beta_\sigma = (-1)^{|\sigma|} \bmod p$, and for a corner outside the cube a *dummy* key with $\beta_\sigma = 0$ at a random point. DPF keys hide β as well as the point, so every query to a cube carries exactly 2^d indistinguishable keys per provider, and a box at the cube's edge cannot be told from an interior one.

Answering. Provider j evaluates its 2^d keys over all M cells, adds them, $e_j = \sum_\sigma e_{\sigma,j}$, and contracts once against the encrypted table; the response is one ciphertext, and the DU decrypts the box aggregate in

Algorithm 1 Outsourcing at the data owner.

Require: records $i \in [N_d]$ with tags, values m_i and roles $\rho(i) \in [R]$; templates $\{w_f\}$; cubes $(\mathcal{F}_1, \ldots, \mathcal{F}_d)$

$(pk, sk) \leftarrow \mathsf{KGen}(1^\lambda, p, \ell)$; $K' \leftarrow \{0,1\}^\lambda$; $D \leftarrow \emptyset$
for all keywords w in arrival order **do**
  if $w \notin D$ **then** $D(w) \leftarrow |D|$ ▷ rows never move
end for
$I[D(w)][i] \leftarrow 1$ for every keyword w of record i
$K'_\rho \leftarrow F_{K'}(\rho)$ for every role ρ
$P[x][c] \leftarrow$ rejection-sampled $F'_{K'_{\rho(c)}}(x, c)$, uniform on $\mathbb{Z}_p$
$I' \leftarrow \mathrm{pack}_{32}(I) + P \bmod p$ ▷ masked, packed index
for all roles ρ and templates f **do**
  $A_f^\rho[x] \leftarrow \sum_{i:\rho(i)=\rho} I[x][i]\, w_f[i]\, m_i$ for $x \in [N]$
  $C_f^\rho[c] \leftarrow \sum_{i:\, c(i)=c,\, \rho(i)=\rho} w_f[i]\, m_i$; $T_f^\rho[c] \leftarrow \sum_{c' \le c} C_f^\rho[c']$
  pack A_f^ρ, T_f^ρ and the record column in slot row 0, with a copy in row 1, and encrypt under pk
end for
return I', ciphertexts and pk to every CP_j; (sk, D, K'_ρ) to the data users of role ρ

Algorithm 2 Verified range query.

Token at the data user, for box B of a registered cube
$\delta_s \leftarrow \mathbb{Z}_p^*$ ▷ fresh per query
for all $\sigma \in \{0,1\}^d$ **do**
  $c \leftarrow \kappa_\sigma(B)$; $\beta \leftarrow (-1)^{|\sigma|}$
  if c lies outside the cube **then** $c \leftarrow$ random cell, $\beta \leftarrow 0$ ▷ dummy
  $(\kappa_{\sigma,1}, \ldots, \kappa_{\sigma,n}) \leftarrow \mathsf{Gen}(1^\lambda, c, \beta)$
  $(\hat\kappa_{\sigma,1}, \ldots, \hat\kappa_{\sigma,n}) \leftarrow \mathsf{Gen}(1^\lambda, c, \delta_s\beta)$
end for
send $\{\kappa_{\sigma,j}, \hat\kappa_{\sigma,j}\}_\sigma$ to CP_j
Answer at provider CP_j
$e_j \leftarrow \sum_\sigma \mathsf{FullEval}(j, \kappa_{\sigma,j})$; $\hat e_j \leftarrow \sum_\sigma \mathsf{FullEval}(j, \hat\kappa_{\sigma,j})$ ▷ $O(2^d M)$ PRG
$\mathrm{ct}_j \leftarrow \boxplus_c E_{j,c} \boxdot T_c$, $E_{j,c}$: chunk c of e_j in row 0, of $\hat e_j$ in row 1
rotate and sum within segments; modulus-switch; **return** ct_j ▷ one ciphertext
Combine at the data user
$(y, \hat y) \leftarrow$ slots $(0, \ell/2)$ of $\mathsf{Dec}\big(sk, \boxplus_j \mathrm{ct}_j\big)$
if $\hat y \neq \delta_s y$ **then abort** ▷ tamper evidence
return y

slot 0 (Algorithm 2; a single-keyword aggregate is the special case $d = 0$, one key pair at α_w over A_f). The cost per provider is 2^d full-domain evaluations over M cells, $\lceil kM/(\ell/2) \rceil$ plaintext-ciphertext products for k templates and $\log_2(\ell/2k)$ rotations; the uplink is 2^d keys of $O(\lambda \log M)$ bits. Nothing depends on the box or on N_d; Lemma 1 requires the cube's total to stay below p.

Why prefix sums rather than interval keys. Function secret sharing for d-dimensional intervals [20] needs the servers to hold each record's coordinates, but over owner-encrypted records the coordinates (which device, which minute, which band) are the sensitive part. A dense, encrypted table over all M cells keeps them hidden and lets each of the 2^d pieces be a point function; the price is that cubes are registered and precomputed, at a cost that grows with M rather than with the record count.

4.6. Tamper evidence

Every response carries a second slot row, so a check value travels with the answer at no extra downlink. A single stored MAC key, row 1 holding δm_i, catches inconsistent responses but not a provider that applies the same substitution to both rows, for instance contracting against $k\, e_j$, which shifts value and tag in the ratio δ. Per-position keys $\delta_x = F''_{K''}(x)$ catch substitutions spread over other positions but not one concentrated on the queried position, and do not exist for contractions over records.

A per-query secret. The session layout stores no key at all: row 1 of every column and table is a copy of row 0. With each query the DU draws a fresh δ_s uniformly from $\mathbb{Z}_p^*$ and sends a second key set for the same points with outputs multiplied by δ_s ($\delta_s\beta_\sigma$ at each corner of a range, fresh shares of $\delta_s v$ in a flexible round).

Each provider places the second evaluation vector in row 1 of the same plaintexts, so the homomorphic work is unchanged and slot $\ell/2$ decrypts to $\delta_s y$; the DU accepts only if $\hat{y} = \delta_s y$.

Theorem 2 (Tamper evidence). *Consider one query in the session layout. Let a coalition of at most $n-1$ providers return arbitrary responses computed from its members' views, with at least one honest provider, an honest DU that keeps δ_s secret, and no provider colluding with the DU. The DU accepts an output $y' \neq y$, where y is the honest answer, with probability at most*

$$\frac{1}{p-1} + \varepsilon,$$

where $\varepsilon = \kappa \operatorname{Adv}^{\mathrm{dpf}}$ for a query with κ DPF keys per provider in its δ_s-carrying set ($\kappa = 1$ for a keyword or a retrieval, 2^d for a range) and $\varepsilon = \operatorname{Adv}^{\mathrm{prf}}$ for a flexible round. Over Q queries the bounds add.

Proof (Proof sketch). Let (r_0, r_1) be what the coalition's responses decrypt to in the two slots and $(y_C, \hat{y}_C)$ what its members would have contributed honestly; all are functions of the coalition's view V. The DU decrypts $y' = y - y_C + r_0$ and $\hat{y}' = \delta_s y - \hat{y}_C + r_1$ and accepts if and only if $r_1 - \hat{y}_C = \delta_s(y' - y)$. Replacing the coalition's δ_s-carrying keys by keys for an independent scalar changes the probability of this event by at most $\kappa \operatorname{Adv}^{\mathrm{dpf}}$, since DPF privacy hides the output as well as the point. Then V is independent of δ_s, and since $y' - y \neq 0$ at most one of the $p-1$ equally likely values of δ_s satisfies the equation. Each query draws a fresh δ_s, so a union bound covers Q adaptive queries (full proof in Supplementary S2). □

The bound puts no restriction on how a provider deviates: it may rescale the queried point, swap or mix slot rows, replay an old response or return any ciphertext. It does not guarantee output delivery, and a DU that reveals its aborts can leak up to one bit about the data per query, which retrying silently through another provider set avoids.

4.7. Retrieval, flexible queries and role-scoped access

Retrieval. Providers return $b_j = \sum_x e_j[x]\, I'[x]$; the DU adds the shares, subtracts $P[\alpha_w]$, which it derives from its mask key, and unpacks the posting list. The response is $\lceil N_d/32 \rceil$ words whatever $|S|$, and payloads are fetched with a batch of DPF retrievals padded to a public bound.

Flexible queries. For a conjunction, a disjunction or weights chosen at query time, the DU first retrieves the posting lists, forms the selection vector v over the records, deals it as fresh additive shares (seeds to $n-1$ providers, an explicit vector to one) and has each provider contract (1) against the record column. Every share a coalition holds is uniform or pseudorandom, so the second round reveals nothing about v; the DU learns S, which it already knows.

Role-scoped access. DPF privacy hides the queried row, so no provider can check *which* keyword or range a query targets, and a user who sends arbitrary share vectors can read every linear function of the columns it reaches. We therefore restrict what the columns and keys cover. The DO assigns each record a role $\rho(i)$, lays out record identifiers so that each packed word holds records of one role, derives $K'_\rho = F_{K'}(\rho)$ and masks word c under $K'_{\rho(c)}$, and builds A^ρ_f, T^ρ_f and the record column from role ρ's records only. A provider answers a role-ρ user from role ρ's columns, selected by the user's authenticated identity; it needs no exchange with other providers.

Proposition 2 (Role confinement). *If every provider authenticates the user's role and applies that role's columns on every path, and no provider colludes with the user, then the view of a role-ρ user, for any queries including malformed keys and arbitrary share vectors, is simulatable from role ρ's records and the public dimensions, up to* $\operatorname{Adv}^{\mathrm{prf}}$.

Proof. Aggregate, range and flexible responses are linear in role-ρ columns. Retrieval returns linear combinations of masked words; words of other roles carry masks under keys $K'_{\rho'}$ the user lacks, which become one-time pads once F is replaced by a random function. □

The mechanism costs storage per role rather than interaction, reveals each query's role to the providers, and does not express rights finer than a record partition.

5. Security analysis

Leakage. The setup leakage $\mathcal{L}_{\text{Setup}}$ is what a provider sees in its replica: (n, t, N, N_d, p, ℓ), the registered templates and ciphertext shapes, the cubes' dimensions $(s_1, \ldots, s_d)$, the number of roles and their spans of packed words, and tier sizes. The per-query leakage to the providers is $\mathcal{L}^{\text{CP}}_{\text{Query}} = \text{class}(q)$: whether a query is a keyword aggregate, a range over a named cube, a retrieval or a flexible round, the user's role, and whether it carries the verification key set. It contains no keyword, range, access pattern, result size or result, and by Theorem 1 the setup leakage contains no posting-list length.

Definition 1 ($\mathcal{L}$-security against t-collusion). In **Real**, the adversary picks a database and a coalition C with $|C| \le t$, receives the coalition's stored state, and adaptively issues Q queries, receiving after each the coalition's keys or shares and the responses those providers compute. In **Ideal**, a simulator produces the same views from $\mathcal{L}_{\text{Setup}}$ and $\text{class}(q)$ alone. CR-SHE is $\mathcal{L}$-adaptively-secure if for every PPT adversary some PPT simulator makes the two experiments indistinguishable up to $\mathsf{negl}(\lambda)$.

Theorem 3 (Correctness). *For templates and cubes satisfying the magnitude bound of Lemma 1, a keyword aggregate returns $\sum_{i\in S} w_f[i]\, m_i$, a range query on box B returns $\sum_{i:\, c(i)\in B} w_f[i]\, m_i$, a retrieval returns exactly S, and a flexible round on v returns $\sum_i v[i]\, m_i$, except with the negligible probability that decryption fails; honest responses always pass the check.*

Proof (Proof sketch). DPF correctness gives $\sum_j e_j = u_{\alpha_w}$ for a keyword and $\sum_\sigma \beta_\sigma u_{\kappa_\sigma}$ for a range, so by Lemma 1 the sum of responses decrypts to $A_f[\alpha_w]$, or to $\sum_\sigma \beta_\sigma T_f[\kappa_\sigma]$, which (3) turns into the box aggregate; the true value lies in $[0, p)$, so the reduction modulo p loses nothing although individual signed terms are negative. Rotations act within slot rows, so templates and check values do not mix, and row 1 decrypts to δ_s times row 0 for honest providers (Supplementary S3). □

Theorem 4 (t-collusion leakage security). *If F and F' are secure PRFs, $(\mathsf{Gen}, \mathsf{Eval})$ is an $(n-1)$-private DPF and the homomorphic and symmetric schemes are IND-CPA, then CR-SHE is $\mathcal{L}$-adaptively-secure against $t = n - 1$ colluding providers, with*

$$\big|\Pr[\mathbf{Real} = 1] - \Pr[\mathbf{Ideal} = 1]\big| \le \text{Adv}^{\text{prf}}_{F'} + Q\big(2^{d_{\max}+1}\text{Adv}^{\text{dpf}} + 2\,\text{Adv}^{\text{prf}}\big) + m\,\text{Adv}^{\text{cpa}},$$

where $d_{\max}$ is the largest registered cube dimension and m the number of ciphertexts in the coalition's view.

Proof (Proof sketch). A sequence of hybrids replaces (1) F' by a random function, after which the masked index is a one-time pad of the all-zero index (Theorem 1); (2) every DPF key the coalition holds, dummies and δ_s-carrying keys included, by simulated keys, which hide both point and output; (3) flexible shares by uniform vectors; and (4) every stored ciphertext by an encryption of 0. Each response is a deterministic function of the provider's keys or shares, the query class and its stored state, so the simulator recomputes it. The final hybrid depends only on the leakage (Supplementary S4). As is standard for BFV, IND-CPA is assumed to hold given the rotation keys. □

Recovering S and then asking providers to aggregate over it would put S itself, a function of the query and the database, in the coalition's view, which no simulator given data-independent leakage can produce. What makes Theorem 4 hold is contracting a share at the provider, not the number of rounds. Because providers never interact and receive only keys or shares, a deviating coalition sees the same view, so deviation buys it no information about the query, and Theorem 2 makes any answer-changing deviation evident.

6. Implementation and experimental setup

Two implementations share the parameters below. A C++17 prototype implements the single-keyword path; it produced the single-keyword measurements on the real datasets. An open reference implementation (Python driving OpenFHE's C++ kernels) implements the full construction: range aggregation in the *combined* (one key set, no check), *verified* (session check) and *per-corner* (per-cell keys) layouts, all three verification modes on every path, role-scoped columns and mask keys, the packed masked index with

rejection-sampled masks, retrieval, the flexible path, and a single-server baseline. Its 83 tests include DPF checks at every point of every domain up to depth 8, range answers against brute force for $d \leq 3$ and $n \leq 3$, and failure controls that must and do fail, such as a flipped corner sign or a provider told δ_s.

Parameters. The two-party DPF is the Boyle–Gilboa–Ishai GGM-tree construction [20] with fixed-key AES-128. The homomorphic layer is leveled BFV in OpenFHE [47] at plaintext modulus $p = 68{,}724{,}326{,}401$, a 37-bit prime with $p \equiv 1 \bmod 2^{17}$, ring dimension 8,192, multiplicative depth 1, at the 128-bit security level. In both implementations the DPF output group and the plaintext ring are one modulus object, so the error Lemma 1 warns about cannot be written.

Datasets. The Environmental Sensor Telemetry dataset [48] has 405,184 messages from three sensor arrays, indexed under device, one-minute window, decile bands of five sensor channels and two flags: $N = 11{,}580$ tags over 3,646,656 postings. The UCI Beijing Multi-Site Air-Quality dataset [49, 50] has 420,370 hourly records from twelve sites after dropping those without a temperature, $N = 35{,}162$ tags over 4,624,070 postings. Both aggregate temperature in hundredths of a degree. Their posting lists are heavy-tailed (Fig. 2): most tags are time buckets holding tens of records (median 35 and 12), while device, site, flag and band tags hold up to 404,702 and 98,878 records, which is what makes posting-list lengths identifying (Section 7.5).

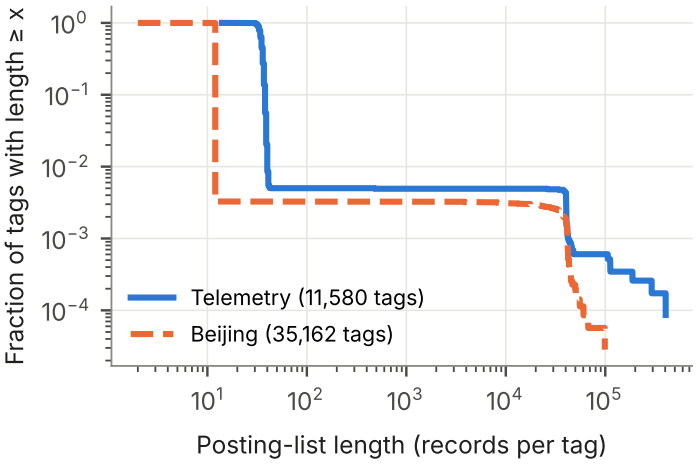


Figure 2: Posting-list length distribution of the two datasets. Plotted with an AI-assisted script.

Machines. C++: an AMD EPYC 7J13 with 64 cores, medians of 11 runs. Reference implementation: synthetic-data runs on a cloud VM with four vCPUs of an Intel Xeon at 2.1 GHz; real-dataset runs on an AMD EPYC 7J13 server with 32 vCPUs; one thread, medians of 5–21 runs. Every table and figure is generated from the result files by a script, and latencies from different machines are not compared with each other.

7. Results and discussion

7.1. Correctness

On both real datasets the reference implementation matched the plaintext ground truth for 200 random tags at each $n \in \{2, 3, 4\}$ on the keyword, retrieval and flexible paths, all 1,800 queries per dataset correct, with a 132,081 B response throughout, and every range query below decrypted to the plaintext box aggregate. The C++ prototype matched the ground truth for 1,000 random tags per template and $n \in \{2, 3, 4\}$ on both datasets and sent a byte-identical response for matched sets of size 0 and 4,999.

7.2. Range aggregation on real data

We built two cubes from the records themselves: device × one-minute window × CO band over the telemetry data ($M = 345{,}690$ cells) and site × day × $PM_{2.5}$ band over the Beijing data, with missing

readings as an eleventh band ($M = 192{,}852$). A query fixes one device or site, a window of L minutes or days and bands 3–7, so $d = 3$ and each provider receives 8 keys, or 16 verified. Fig. 3 breaks the per-provider cost into DPF evaluation and the homomorphic part.

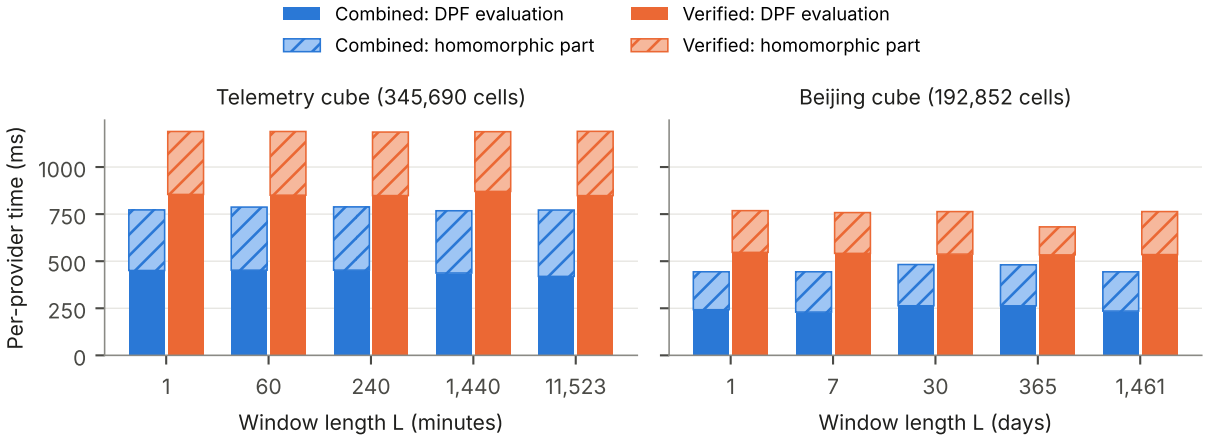


Figure 3: Per-provider range-query cost on the real cubes. Plotted with an AI-assisted script.

Cost does not depend on the range. A verified query costs 1.19 s per provider on the telemetry cube and 0.68–0.77 s on the Beijing cube, and a combined query 0.77–0.79 s and 0.44–0.48 s, with no trend in L from one window to the whole timeline. DPF evaluation is the larger share, because every key is expanded over all M cells, and verification adds a second key set rather than more homomorphic work. A box at the cube's origin, with seven of its eight corners dummies, costs the same (1,192.6 and 758.9 ms verified). Every response is one 132,081 B ciphertext, and the data user spends 1.1–1.9 ms combining, checking and decrypting.

Verification costs one key set, not a layout. The verified layout stores the same ciphertexts as the combined one and performs the same homomorphic products. On a synthetic cube of the telemetry dimensions it costs 1.6–1.9 times the combined layout and less than half the per-corner layout (3.24–3.44 s), which stores the table eight times and still misses deviations at the queried corner (Fig. 4, left). Range cost grows linearly with the cube: 34.7, 129.3 and 1,356.0 ms per provider for 10^4, about 10^5 and 10^6 cells (right), so cube size, not record count or range length, is what a deployment provisions.

7.3. Tamper evidence

Fig. 5 injects five provider deviations, 200 trials each, into every path and verification mode; one provider deviates and the other is honest. The deviations add a constant to the value slot, a uniform vector to the share, scale the share, perturb a random point, and perturb exactly the point the query reads, as if the provider knew the query. A single stored key catches only the inconsistent constant: every consistent substitution was accepted with a wrong answer on every path. Per-position keys catch every deviation except the one aimed at the queried point, which passed 200 times out of 200 on the keyword and range paths. The session check detected every deviation that changed the answer: across the 4,000 session trials no wrong answer was accepted, and the undetected keyword trials perturbed an empty row, so their answers were correct. On both real cubes it also flagged all 250 tampered range queries, while the combined layout without a check returned 200 of them wrong and unflagged.

7.4. Single-keyword aggregates and baselines

In the C++ prototype a keyword aggregate costs 38.8–39.2 ms per provider across a 405× range of record counts, because the precomputed column removes N_d from the query, while a single-round variant that contracts the masked index row by row grows to 1,378.2 ms at the full dataset (Fig. 6). Table 3 places

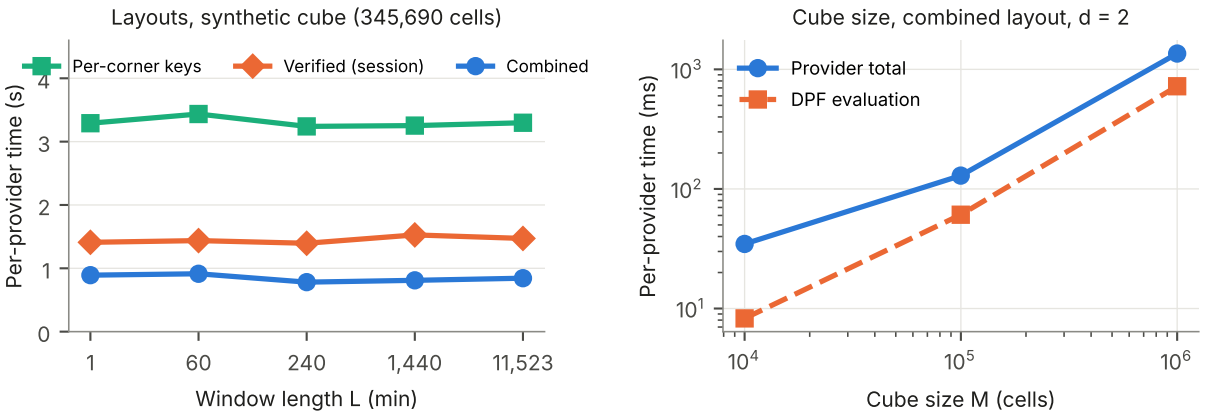


Figure 4: Range layouts against window length (left) and cost against cube size (right). Plotted with an AI-assisted script.

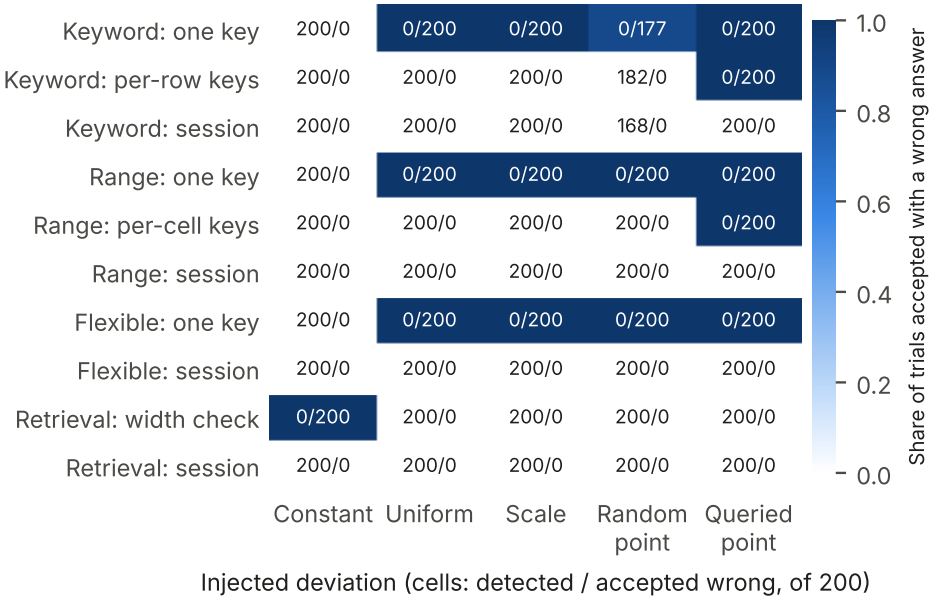


Figure 5: Injected provider deviations: detected / accepted with a wrong answer, of 200 trials. Plotted with an AI-assisted script.

the keyword aggregate against baselines for a tag with the median $|S| = 35$. The DORY-style search-only baseline costs 3.3× more because it materialises the selection vector; a faithful homomorphic equality scan needs depth 17 and costs 617× more, consistent with the fifteen hours HADES reports for a million encrypted records.

Packing and the single-server alternative. In the reference implementation the packed response halves provider work and downlink: a single template falls from 53.0 to 23.5 ms at $N = 11{,}580$, and four templates fit in one compressed 132,081 B ciphertext in 32.7 ms, against 194.9 ms and 2,104,088 B for four unpacked responses (Fig. 7, left). A single server holding the same columns costs 42.6 and 92.5 ms at the two tag domains, 1.8 and 2.9 times one CR-SHE provider, but two providers together spend 47.0 and 62.8 ms, so in total server work the trust split is not cheaper. Its case rests on the client (right): the single server's query is

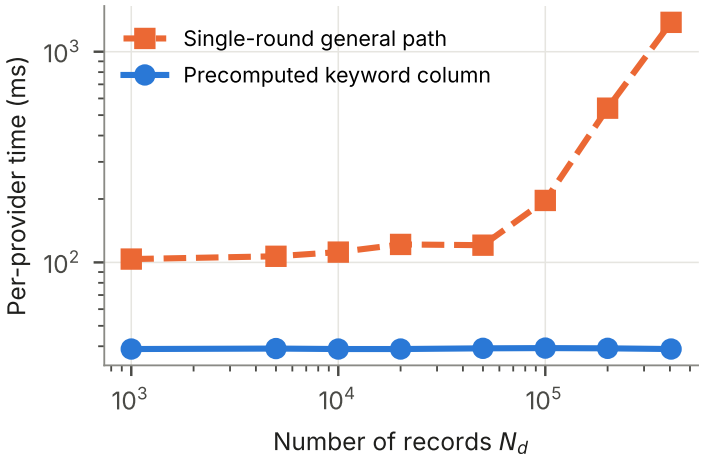


Figure 6: Keyword aggregate against record count, telemetry data, C++ prototype. Plotted with an AI-assisted script.

Table 3
Keyword aggregate against baselines ($N_d = 10^5$, C++; † extrapolated).

System	Server (ms)	Downlink (B)	Leakage to the server(s)
Plaintext scan	< 0.001	8	everything
SSE, 1 server	< 0.001	560	search and access pattern, $\lvert S\rvert$
Path ORAM	$0.059^{\dagger}$	373,248	$\lvert S\rvert$, via access count
DORY-style search	130.4	800,000	none; no analytics
FHE equality scan	$24{,}157^{\dagger}$	n/a	none; 1 server
CR-SHE, keyword	**39.2**	**525,950**	$\mathrm{class}(q)$ only

789,033 and 2,367,099 B of fresh ciphertexts against a 260 or 294 B DPF key per provider, and for the range cubes 44.4 and 25.0 MB (computed from the measured ciphertext size) against 5.5 and 5.2 kB of keys, three to four orders of magnitude, the regime of a thin or metered IoT client.

7.5. Masked index: cost and leakage

Cost. Packing 32 membership bits per word divides the masked index by exactly 32 and makes retrieval 19 to 27 times faster at $N_d = 10^4$ and 2×10^4, where both layouts fit in memory (Fig. 8). At the full 405,184 records the packed index is 1.17 GB per provider, against 37.5 GB unpacked (computed), and a retrieval takes 1.10 s and returns 101,296 B.

Leakage. We mount a frequency and volume leakage-abuse attack [9, 10] against four representations of the index: unmasked, whose rows expose every posting-list length; padded to the next power of two; volume-hiding, every row padded to the longest; and the masked index (2), with row statistics computed under the real PRF. The adversary's auxiliary knowledge is the tag frequencies of a similar deployment, modelled by thinning the true lengths to a fraction f of the records; it assigns each row to the tag of nearest auxiliary frequency, breaking ties at random (Fig. 9, Table 4).

Unmasked, the index gives away the tags that carry meaning: every device, flag, temperature band and wind direction, and 92.5–100% of most band families. At $f = 1$ a tag is missed only when another tag has exactly the same length, and the attack then reaches the best any length-only attacker can do: 0.484% and 0.324% overall, 56× and 114× random guessing, low only because time buckets of near-identical length dominate the tag universe. *Power-of-two padding*, at 1.55× and 1.54× storage, still re-identifies 9× and 11× random guessing, and 75% of the motion flags. *Volume-hiding padding and the masked index* both reduce the attack to random guessing at every f on both datasets, but volume hiding inflates postings 1,285× and 752×, while the packed masked index costs 2× a plaintext bitmap. In the masked column the attack's

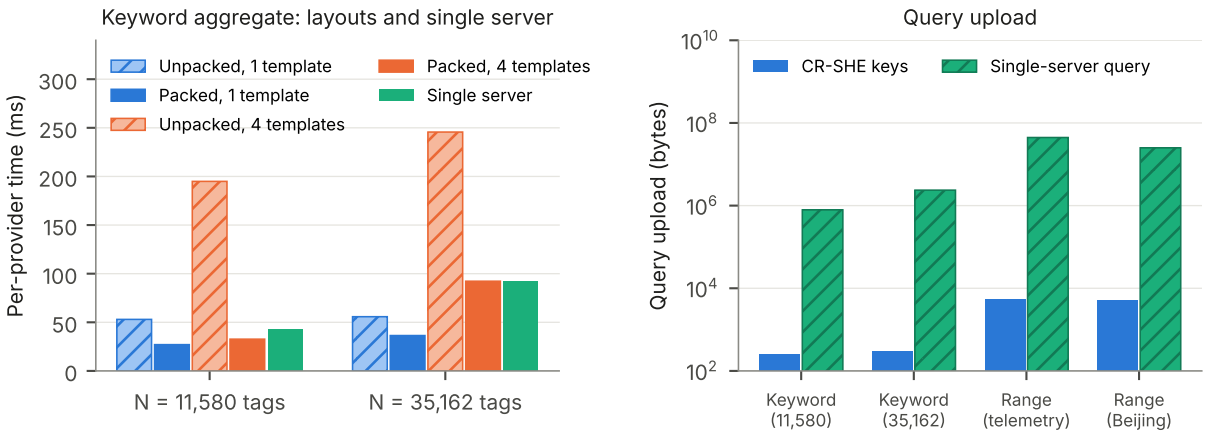


Figure 7: Keyword-aggregate cost by response layout (left) and query upload against a single server (right). Plotted with an AI-assisted script.

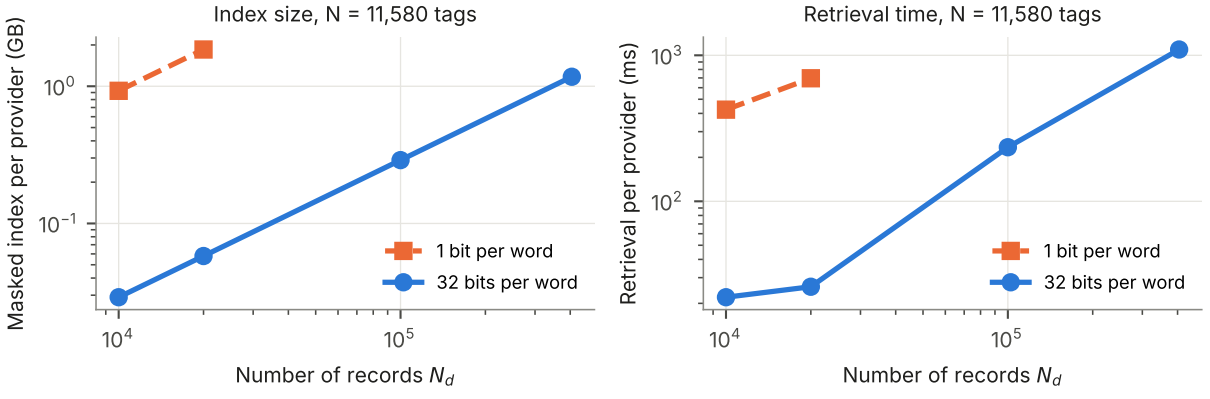


Figure 8: Masked-index size and retrieval time, packed and unpacked. Plotted with an AI-assisted script.

guesses all fall on the longest posting list, so one tag per dataset is matched by default. Higher figures from an earlier run in our C++ prototype exceed this length-only bound and are withdrawn (Supplementary S6).

7.6. Role-scoped access and deployment

A role-0 user sending raw shares for all 512 rows of an index over three roles of 4,096 records recovered every row, including all 365,227 membership bits of the other roles, when one mask key was used, and none of those bits with role-scoped keys; 6.06% of the other roles' words passed the width check under its own key, against the 6.25% expected of uniform words. With $n \in \{2, 3, 4\}$ provider processes over loopback, provider compute was flat in n, 34.82–35.48 ms, and end-to-end latency rose from 44.1 to 52.5 ms in transport alone.

7.7. Discussion

Ranges are where the design pays. A registered range aggregate costs the same for a one-minute window as for a four-year timeline and returns one ciphertext, where single-tag aggregates need one query per cell and a single server an encrypted query as long as the cube. The per-query secret removes the old choice between a cheap unverifiable layout and an expensive per-corner one. *Relation to prior art.* The 2^d box decomposition

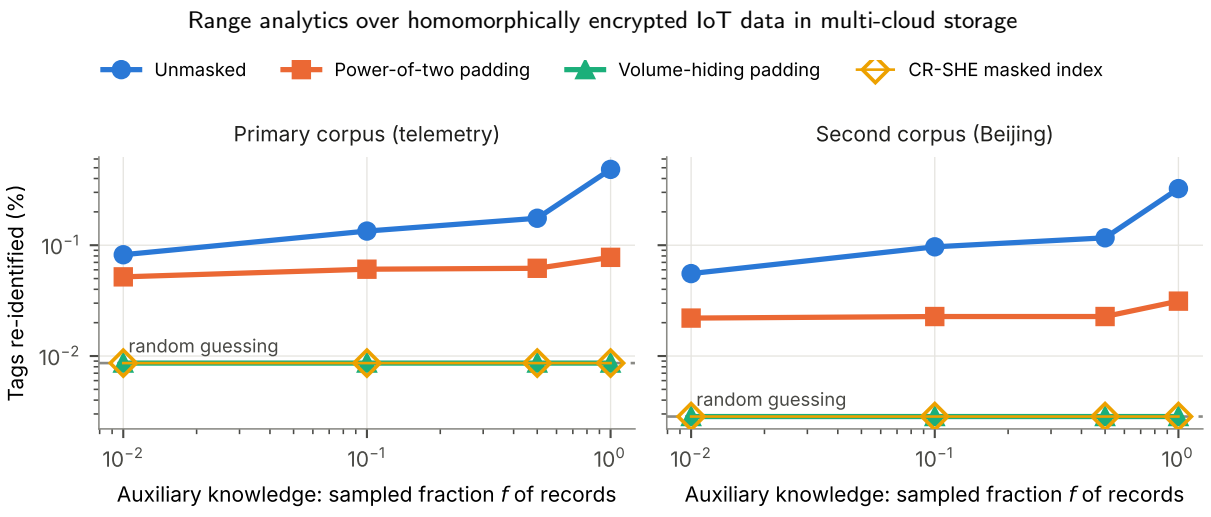


Figure 9: Tags re-identified against the adversary's auxiliary knowledge. Plotted with an AI-assisted script.

Table 4
Tag re-identification by class at $f = 1$.

Tag class	Unmasked	Power of two	Volume-hiding	Masked
Primary corpus				
Device	100.0%	55.6%	0.0%	0.0%
Light flag	100.0%	41.7%	0.0%	0.0%
Motion flag	100.0%	75.0%	0.0%	50.0%
CO band	32.5%	2.0%	0.0%	0.0%
LPG band	32.5%	2.0%	0.0%	0.0%
Smoke band	32.5%	2.0%	0.0%	0.0%
Humidity band	92.5%	2.0%	0.0%	0.0%
Temperature band	100.0%	11.8%	0.0%	0.0%
One-minute window	0.2%	0.0%	0.0%	0.0%
All tags	0.484%	0.078%	0.009%	0.009%
Random guessing		0.0086%		
Second corpus				
Site	33.3%	1.2%	0.0%	0.0%
Wind direction	100.0%	10.4%	0.0%	0.0%
$PM_{2.5}$ band	95.5%	3.3%	0.0%	0.0%
PM_{10} band	100.0%	5.6%	0.0%	0.0%
SO_2 band	100.0%	18.3%	0.0%	10.0%
NO_2 band	100.0%	3.3%	0.0%	0.0%
CO band	100.0%	3.0%	0.0%	0.0%
O_3 band	100.0%	12.2%	0.0%	0.0%
Pressure band	100.0%	10.2%	0.0%	0.0%
Dew-point band	95.5%	1.7%	0.0%	0.0%
Hour	0.0%	0.0%	0.0%	0.0%
All tags	0.324%	0.031%	0.003%	0.003%
Random guessing		0.0028%		

[20], the one-time-padded index [13] and the client-chosen one-time MAC key [29] each have a precursor; our contribution is their composition into one-round, verifiable range analytics over owner-encrypted replicas at $t = n - 1$, which none of the systems in Table 1 provides.

8. Conclusion

IoT analytics are range aggregates over time, place and value, and the data behind them increasingly sit encrypted across several clouds. CR-SHE answers such aggregates in one round over data encrypted under one owner's key, hiding the range, the matched records, their number and the result from any coalition of up to $n-1$ clouds that never communicate. Encrypted summed-area tables turn any box into 2^d padded point keys, oblivious homomorphic aggregation keeps the matched set from ever being assembled, a secret scalar drawn per query makes tampering evident at no extra homomorphic cost, and role-scoped columns confine each user to its own records. On two real IoT datasets a verified range query costs 0.68–1.19 s per cloud with a fixed 132 kB response and kilobytes of upload, no tampered answer was accepted, and a frequency attack fell to random guessing where power-of-two padding did not.

The work has two main limitations. First, range aggregation, the session check and role-scoped access were measured in a single-threaded reference implementation with providers on one host, so the absolute latencies are those of an unoptimised driver and wide-area effects were not measured. Second, one-round ranges apply only to cubes registered in advance whose totals stay below the plaintext modulus, and privacy is proved for semi-honest providers. Future work will port the range path to native code, evaluate cross-region deployments, add succinct multi-party DPFs for $n > 2$ and support multiple data owners.

Supplementary material

The Supplementary Material (PDF) accompanies this article: S1, the full argument for a homomorphic layer at $t = n - 1$; S2, the stored-key verification modes and the full proof of Theorem 2; S3 and S4, the full proofs of Theorems 3 and 4; S5, dynamic ingest, the complexity table and further measurements (range layouts and cube size, selection latency, packing and the single-server baseline, the flexible path, the packed index, deployment, communication, setup and scaling, role extraction and the second dataset in the C++ prototype); and S6, the withdrawn leakage figures of an earlier C++ run.

CRediT authorship contribution statement

Md Mazharul Islam: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Writing – original draft, Writing – review and editing. **Mubasshir Ahmed:** Software, Data curation, Investigation, Writing – review and editing. **Md Redowan Zaman Anik:** Software, Data curation, Visualization, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The two datasets are public: the Environmental Sensor Telemetry Data on Kaggle [48] and the Beijing Multi-Site Air-Quality Data in the UCI Machine Learning Repository [50]. The reference implementation, benchmark scripts and result files are not yet public; they will be made available by the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used three AI assistants. Claude (Anthropic) was used to develop the reference implementation with its tests and benchmark scripts, write the scripts that draw all figures, search and compare the literature, draft and edit parts of the text and analysis, including the range construction, the tamper-evidence mechanism, role-scoped access and their proofs, condense the manuscript to its present length, and convert it to the journal's format. ChatGPT (OpenAI) and Gemini (Google) were used to edit and review intermediate drafts; changes adopted from them were checked against the implementation and the measured results. No image was generated directly by a generative AI model: Fig. 1 is a schematic drawn by a script, and every other figure is plotted by a script directly from the measured results. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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